

7.4 Gene technology

- i Understand how recombinant DNA can be produced, including the role of restriction endonucleases and DNA ligase.
- ii Understand how recombinant DNA can be inserted into other cells, and the use of various vectors such as viruses and gene guns.
- iii Understand how antibiotic resistance marker genes and replica plating are used to identify recombinant cells.
- iv Understand how 'knockout' mice can be used as a valuable animal model to investigate gene function.
- v Understand the process of genetic modification of soya beans and how it has been used to improve production, including altering the balance of fatty acids to prevent oxidation of soya products.
- vi Understand why the widespread use of genetic modification of major commercial crops and other transgenic processes have caused public debate of their advantages and disadvantages.